

Project Case Studies

Representative applications for tunnels, rail, dams, and construction support programs.

These summaries mirror the sectors highlighted across the site and show how GIM Nextgen solutions are positioned for real-world monitoring challenges.

Tunnel convergence monitoring

Challenge: Excavation teams needed continuous visibility into deformation while advancing through variable ground conditions.

Solution: Wireless tilt and displacement instrumentation was paired with a gateway and cloud dashboard for live convergence review.

Outcome: Engineers received earlier warning of movement trends and validated lining performance throughout excavation stages.

Rail track stability monitoring

Challenge: A rail corridor required close tracking of settlement and cant to maintain safe operations during nearby works.

Solution: Settlement markers, tilt instrumentation, and automated reporting created a clear picture of vertical and lateral movement.

Outcome: The operator gained faster review cycles and better evidence for maintenance and possession planning.

Dam safety instrumentation

Challenge: The owner needed a practical way to review pore pressure, seepage indicators, and long-term structural movement.

Solution: Piezometers, deformation sensors, and remote dashboards were organized into a single monitoring workflow.

Outcome: Annual safety reviews were supported by cleaner trend history, alarm management, and easier reporting access.

Construction support during excavation and piling

Challenge: Contractors required quick feedback to manage risk while sequencing heavy temporary works.

Solution: A site-wide mix of displacement, vibration, and environmental sensors was integrated with real-time alerts.

Outcome: Project teams responded faster to threshold events and maintained a clearer audit trail during critical activities.

Industries served

- Rail and metro infrastructure.
- Tunnels and underground works.
- Dams, reservoirs, and water-retaining assets.
- Buildings, excavation support systems, and major construction sites.

Delivery approach

- Instrumentation selection based on project risk and site conditions.
- Installation and commissioning support with practical handover guidance.
- Remote data review workflows for owners, consultants, and contractors.
- Scalable support from single-site deployments to larger monitoring portfolios.

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